

SIDDARTH BOGGARAPU

Bangalore, India

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Builder portfolio centered on systems that **verify, cite, and refuse** under uncertainty — LLM instrumentation, grounded QA, multi-host security correlation, paper trading research, agent ops, legal-tech RAG, and physical simulation. CSET / SaTML submissions are **under review** (not accepted, not published).

RESEARCH / WORKSHOP SUBMISSIONS (PENDING)

CSET · SaTML — under review

Trust / evaluation / Anima-adjacent materials · awaiting decision

- Do not list as published or accepted until official notice.
- Private anonymized SaTML 2027 System A evaluation artifact prepared for the review process.

VENTURES AND PROFESSIONAL EXPERIENCE

Orqis — Technical cofounder

Live product · FastAPI · libcst · MCP · Redis · human review before merge

- Owns backend for an agent-ops product: detect runaway loops, explain via MCP, open a reviewable patch/PR.
- Deterministic libcst remediations plus confidence-gated LLM assist; never silent-pushes the default branch.
- Live product with beta onboarding — strongest shipping founder credential.

Athera (athera.digital) — Founder

AI automation and websites for small businesses · 6+ months

- Ships SMB automation, lead/outreach workflows, and client sites; real clients (names private).
- Ops glue across sheets, Gmail, webhooks, and live deploys.

VidhiSetu (VidhiSethu / Lexprobe) — Founder / builder

Indian-legal RAG with citation audit · closed beta

- Natural-language legal research over an Indian corpus; citation-audit layer filters unsupported claims.
- Working with law agencies and lawyers; public architecture docs; application remains private.

Evex — Founder / lead, event company (Bangalore)

Operations and P&L

- Hosted 20+ school graduation events across Bangalore; ~\$5L+ profit.
- Leadership, logistics, and client delivery outside pure coding.

INTERNSHIPS

Indian Institute of Science (IISc) — Aerodynamics intern

May 2025 · ~10 days (not a long research residency)

- Lab exposure to vortex-ring formation and instability regimes.
- Follow-on CFD / propeller / VRS work was self-directed (not an IISc appointment).

Vegam Solutions — Engineering intern

Hardware filtration (2025 Q3) · Text-to-SQL RAG (2026 Q1, NDA)

- Hardware: end-to-end water-based air-filtration prototype under shop constraints.
- Software: NL→SQL retrieval with schema awareness and grounded refusal; clean-room portfolio proof: text2sql-rag (Spider).

SELECTED PROJECTS (TECHNICAL)

Case-file order matches the portfolio. Each entry: what it does, what was built, honest limits.

Anima — LLM interpretability meter (flagship)

Public · MIT · HF Spaces demo · linked to CSET/SaTML (under review)

- Forward hooks into Hugging Face causal LMs; probe heads read valence, arousal, and uncertainty per token.
- Guard recommends abstaining when the readout looks unreliable (HaluEval / TruthfulQA fixtures).
- 5-model council: TinyLlama 1.1B aggregate 94.0; GoEmotions valence Pearson $r \approx 0.19$ (reported honestly).
- Technical: FastAPI + WebSocket stream, dashboard, public Spaces demo — meter, not a claim the model feels.

BumbleBee — Transformer-free CEC correlator (private)

Open sourcing soon · no Hugging Face transformer on the product path

- Closed-corpus ask / cite / refuse: GRU local encoder → SoftCorrelator (sparsemax) → CoverageGate → cite or refuse.
- Skills (chat / math / code) short-circuit via packs/tools; hard asks fall back to CEC. Not a frontier next-token LLM.
- Honesty: on the sealed pack BM25 still leads in-domain F1; skills wins ≠ core CEC quality. Patent: nothing filed.
- CLI: ingest, seal, train, calibrate, eval, ask, chat, dash. Footprint-first (~24M params class) — not dense-LLM parity.

Corvex — Multi-host campaign correlator (flagship)

Public · MIT · sealed synthetic eval · claim_allowed=false by default

- Fuses weak per-host detectors (lateral_auth, micro_exfil, recon_fanout) into ATT&CK-shaped campaign timelines.
- Observe-only Windows/macOS sensors; live containment locked off by default (human stays in control).
- Holds on synthetic fleets — not validated on real enterprise telemetry or a pure-benign baseline yet.
- Technical: HMAC-signed EventEnvelopes + transitive correlator merge across hosts.

GodFather — Multi-agent NSE paper desk (private)

Opensourcing soon · paper / session-sim only · not investment advice

- Scout / Risk / Session agents share one desk; Risk may cut size or force cash — never raise risk.
- Prior-session bulk/block BUY filings, open-range entry, same-day exits; CLAIMS.md honesty gates.
- Never sent a live broker order. Cut from the broader trade_bot lab.

Orqis — Agent-ops self-healing (also under Ventures)

Live product · detect → explain → patch → human review

- Incident console for runaway tool loops; reviewable unified-diff / PR; refuse silent push.

VidhiSetu — Indian-legal RAG (also under Ventures)

Citation audit · closed beta · public architecture docs

- Answers grounded in sources; unsupported claims filtered before response ships.

Drift — Clinical rule-based health risk tracker

ACC/AHA · FINDRISC · safety-gated experimental ML

- Deterministic cardiovascular / diabetes risk calculators; ML path labeled experimental — not a medical device.
- Technical: Flask + SQLite; Fitbit / Google Fit optional sync.

GeoQuant — Algorithmic trading research platform

Walk-forward · cost-aware · honesty-first

- MLP signals, news sentiment, paper hooks, retraining; publicly reports poor Sharpe (~-0.47) instead of hiding failure.
- Not investment advice. Technical: PyTorch + FastAPI with costs inside the optimizer.

Aerospace / CFD / drone arc (self-directed)

After IISc spark · simulation + hardware learning

- vortex-tracker (OpenCV diameter + speed) · Propeller-simulator (BEMT sweeps) · Drone-Vortex-Ring-Simulation (Helmholtz / Kelvin I).
- NeuralVortex: early FNO-style volumetric surrogate (smoke-scale — not production accuracy).
- Homemade F450 quad + F550 hex builds; Pico FC learning firmware (props-off until tether tests pass).

Other supporting systems

- trade_bot: NSE bulk-deal ingest, scoring, WhatsApp alerts — lab that fed GodFather.
- text2sql-rag: Spider clean-room with schema linking, few-shot, sqlglot validation.
- FleetControl: multi-host stub LLM memory pools + Anima probes.
- cursor-llm-council: multi-model Cursor council that resists yes-man answers.
- AI-BRAIN: voice-first daily OS (Whisper, local/cloud LLM, Qdrant RAG).
- Homelab: Raspberry Pi NAS + Pi-hole; Elevyx: wound-down real-estate lead recovery.

HARDWARE

- Multitrotors: F450 quad + F550 hex (GPS / autonomous modes) — airframe, ESCs, FC tuning.
- Rovers: Bluetooth, line-follow, ultrasonic avoid; custom PCB plant-watering (soil moisture).
- Arduino / ESP32 toolkit including BLE macrodeck for workflow shortcuts.
- Neural Sync (in progress): low-cost 2-channel EEG research instrument — software-first, never mixes synthetic with live serial.

COMMUNITY · SPORTS

- Websites / volunteering: Project Thrive, Project Lighthouse, Arogi Foundation.
- Football: school team 4 years; CBSE state tournament; Goa Globe; ESSB / GameON.
- Past competitive skating (ForceOne Academy). Motorcycle track training (trained, not certified).
- All-India CBSE 2nd — football, Goa Globe.

Portfolio dossier (live widgets + GitHub index): <https://siddarthb07.github.io/siddarthb/>